Paper id: 250873

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VI) THEORY EXAMINATION 2024-25
TOOL DESIGN

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question
a.	How can a machine tool be defined? What socioeconomic roles are played by machine tools for the industries and the nation?
b.	Draw the single-point cutting tool geometry and show all the angles.
c.	What factors need to be considered while selecting a motor for any machine tool drive?
d.	What is tool chatter?
e.	Differentiate between guideways and slideways?
f.	Why do conventional machine tools provide a large number of spindle speeds?
g.	What are the roles of CNC machines in the defense manufacturing industries?

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

Q no.	Question
a.	What do you mean by tool wear? Explain any three methods of tool wear with a neat sketch.
b.	Explain any five tool materials with their applications used in general-purpose machine tools.
c.	Write the short notes on the following: (i) Lead screw (ii) collet (iii) Indexing
d.	Derive the maximum chip thickness developed in the slab milling operation, considering the rectangular block workpiece.
e.	Explain the following: (i) Borings (ii) Drilling (iii) Reaming

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question
a.	Explain the locating principle used in machine tool design. Explain the 321 principles with a neat sketch and its significance.
b.	What is a gear-cutting operation? How does it perform?

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4. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question
a.	Determine the actual machining time required for drilling operation of a hole 25 mm in a 60mm thick plate at a velocity of 55 m/min and a feed of 0.24 mm/rev by an HSS drill of a cone angle of 120*. Assume the approach and overrun is 2mm.
b.	Derive the optimum machining time required (T_{opt}) $T_{opt}= (\frac{1}{n} - 1) * TCT,$ Where TCT= Tool Changing time , n= Index of tool life

5. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question
a.	What is tool life? How can it be measured?
b.	Determine the actual machining time required for the plane milling of a rectangular surface length of 200 mm and width 50 mm by a helical fluted plain HSS cutter of diameter 70 mm, length 75mm and teeth 6. Assume overrun and approach 5mm, cutting velocity 44 mm/min and feed 0.2 mm/tooth.

6. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question
a.	Draw the lathe machine with a neat sketch. Explain the operations that can be performed on the lathe machine.
b.	Explain how the machine tools are classified.

7. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question
a.	What is the layout of spindle speeds in machine tools? Explain any one method with details.
b.	Explain the roles of Jigs and fixtures in machine tools.

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